

The U.S.–China trade war and the volatility linkages between energy and agricultural commodities

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Abstract

Disruptive changes in the competitive environment, such as the U.S.–China trade war, may influence the price volatility of crude oil and agricultural commodities. This study examines the volatility linkage between crude oil and agricultural commodity markets in the context of the U.S.–China trade war and compares the impact of the trade war with that of other exogenous shocks, such as the global financial crisis and the COVID-19 pandemic and associated recession. The results show that the volatility of soybeans exhibits the highest level of responsiveness to the U.S.–China trade war—unsurprising, given the U.S. agribusiness trade to China is dominated by soybeans—followed by coffee and cotton. The size and the dynamic of the impact are largely commodity-specific. Notably, most agricultural commodities are impacted more extensively by the trade war than by other exogenous shocks. These findings are significant not only for the decision making of investors and portfolio managers but also for countries that import and export commodities because changes in the volatility dynamics of crude oil and agricultural commodities often impact export revenues and import expenditures; these changes consequently feed through exports to the global supply chain during exogenous shocks such as the U.S.–China trade war.

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1. Introduction

Investors and portfolio managers in energy and agricultural commodities optimize their allocation strategies and risk management decisions based on the dynamics of return, volatility, and covariances, while considering the impacts of exogenous shocks (e.g., global event shocks) or commodity-specific shocks on these dynamics, which can be relevant and influential (Hasanov et al., 2016). Empirical evidence suggests that exogenous shocks can stimulate spillover effects on commodity market volatilities (Ahmadi et al., 2016; Bakas & Triantafyllou, 2018; Hasanov et al., 2016), complicating portfolio management. In this regard, the U.S.–China trade war, which started in 2018, represents an exemplary exogenous shock. Triggered by several cumulative and unresolved issues, it has impacted energy and agricultural commodity markets and their international trade; in particular, agricultural commodities represent a significant battleground (Bloomberg News, 2018) given that increased demand for biofuels has increased the demand for the production of biofuels such as soybean, corn, wheat, and sugar. Disruptive changes to the global competitive environment can directly or indirectly impact the volatility of energy and agricultural commodities, including the disruptive changes deriving from exogenous shocks, which have a demand-side effect on energy and agricultural commodity markets (Bloomberg News, 2018; Bakst & Beaumont-Smith, 2018; Holland & Sam, 2019). According to the Congressional Research Service report, the U.S. agricultural sector's trade outlook is vulnerable to international trade disputes due to the ongoing trade war.¹ Figure 1 shows that U.S. exports dropped from \$143.3 billion in 2018 to \$135.7 billion in 2020 due to the ongoing dispute with China.² In fact, China imported only \$16.31 billion of U.S. agricultural products in 2018 compared to an average of \$20.5 billion during the period 2016–2017; the imports further decreased to \$10.09 billion in 2019, before rebounding to \$17.04 in 2020.³

¹ https://www.everycrsreport.com/files/20191218_R46132_1235ee78c6fc289df14b59d29ee7800fb2fbefee.pdf

² <https://www.statista.com/statistics/220767/total-value-of-us-agricultural-exports-since-2000/>

³ <https://www.statista.com/statistics/279003/value-of-us-agricultural-exports-to-china-since-1990/>

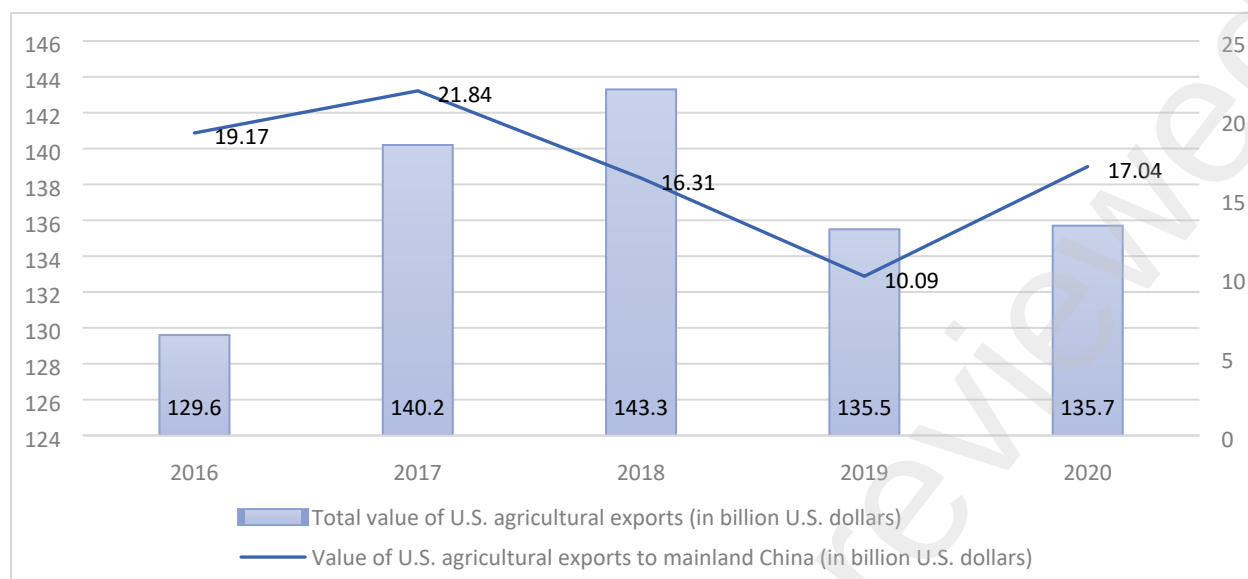


Fig. 1. The total value of the U.S. agricultural exports and value of U.S. agricultural exports to China (in billions of U.S. dollars)⁴

Interestingly, China is one of the United States' largest agricultural export markets, and the United States is the third-largest supplier of agricultural imports to China (Jiang, 2020), meaning U.S. farmers and agricultural firms rely heavily on export markets to sustain prices and revenues because agriculture productivity grows faster than domestic food demand (Bakst & Beaumont-Smith, 2018). Similarly, in the United States, Chinese imports for some agricultural commodity markets are crucial because consumer demand exceeds the domestic production capabilities. The export and import activities between countries benefit consumers by providing lower prices and more product choices. However, tariffs imposed on energy and agricultural commodities increase production costs and food prices (Bakst & Beaumont-Smith, 2018; Holland & Sam, 2019).

Price uncertainty due to structural shocks can precipitate greater uncertainty surrounding oil prices (Herrera et al., 2019). Structural shocks, such as the prolonged U.S.–China trade dispute, can impact investment spending, employment and business sentiment,⁵ and adversely affect investor confidence (Vilmi et al., 2019). As a crude oil exporting country, the United States' share of total global crude oil production (16.2%) was below that of the Organization of Petroleum

⁴ <https://www.statista.com/statistics/279003/value-of-us-agricultural-exports-to-china-since-1990/>;
<https://www.statista.com/statistics/220767/total-value-of-us-agricultural-exports-since-2000/>

⁵ <https://www.scmp.com/news/china/diplomacy-defence/article/2160932/us-concern-about-impact-china-trade-war-jobs-and-prices>

Exporting Countries (41.5%) in 2018 (Dudley, 2019), demonstrating how the trade war has increased uncertainty surrounding future oil prices and indicating challenges for the U.S. government in terms of stabilizing crude oil prices. This may adversely affect investor confidence (Vilmi et al., 2019), thus causing greater volatility in energy and agricultural commodity markets.

Previous studies show that crude oil and agricultural commodity markets are interrelated. For example, Yahya et al. (2019) note that the price of crude oil relates to the demand for biofuels production, the production cost of agricultural commodities, and the rapid economic expansion of developing and emerging countries. According to Huang et al. (2012), low energy prices affect the value of biodiesel production through the reduction in the demand for biofuels. The tariffs on energy and agricultural commodities increase production costs and food prices (Bakst & Beaumont-Smith, 2018; Holland & Sam, 2019) and make it difficult for the United States to seek alternate markets and for China to obtain another supplier (Nag & Ayyub, 2018). Furthermore, the retaliatory tariffs imposed by China have negatively affected the U.S. farm sector. Specifically, China imposed tariffs on more than 90% of U.S. agricultural exports, effective July 6, 2018 (Mor, 2018), and within a month, the estimated prices for U.S. soybean and corn reduced 3%–4% and 5%–6%, respectively.⁶

In the context of the relationship between oil and agricultural commodities, many researchers explore the relationship between energy and agricultural commodities. They employ a wide range of statistical methods, such as cointegration and VECM (Allen et al., 2018; Fowowe, 2016; Natanelov et al., 2011; Pal & Mitra, 2017; Saghaian, 2010; Serra et al., 2011), a copula (Ji et al., 2018; Mensi, Hammoudeh, et al. 2017; Reboredo, 2012), and combination of wavelets and copula (Mensi, Tiwari, et al., 2017), linear and nonlinear Granger causality (Fowowe, 2016; Nazlioglu & Soytas, 2011, 2012; Pal & Mitra, 2017; Su et al., 2019), structural VAR (Fernandez-Perez et al., 2016; Wang et al., 2014), multivariate GARCH (Abdelradi & Serra, 2015; Al-Maadid et al., 2017; Cabrera & Schulz, 2016; Du & McPhail, 2012; Fernandez-Daiz & Morley, 2019; Gardebroek & Hernandez, 2013; Hasanov et al., 2016; Mensi et al., 2014; Pal & Mitra, 2019; Serletis & Xu, 2019), multivariate heteroscedastic autoregressive (HAR) model combined with the DCC-GARCH model (Luo & Ji, 2018), and quantile regression (Algieri & Leccadito, 2017).

⁶ Based on the statistics from the U.S. Farm Bureau (<https://www.fb.org/news/farm-bureau-details-trade-tariff-impacts-on-agriculture>).

Furthermore, previous studies on the relationship between energy and agricultural commodities have mostly sought to identify causes and effects, especially following the 2008 global financial crisis (GFC) (Chen et al., 2010; Du et al., 2011; Su et al., 2019). Nonetheless, there have also been investigations into significant volatility linkages between energy and agricultural commodity prices (Ahmadi et al., 2016; Cabrera & Schulz, 2016; Fasanya & Akinbowale, 2019; Hasanov et al., 2016; Luo & Ji, 2018) and significant spillover effects between energy and agricultural commodities prices in the contexts of the 2006 food crisis and the GFC (Al-Maadid et al., 2017; Kang et al., 2007). Although the U.S.–China trade war has considerably impacted economic activity, this exogenous shock's influence on energy and agricultural commodity price volatility remains understudied. Specifically, the literature has not focused on how and to what extent the U.S.–China trade war influences the conditional volatility linkages of energy and agricultural commodities.

In this study, we address this research gap by investigating the conditional volatilities of energy and agricultural commodities and their interlinkages in the context of the U.S.–China trade war. Our analysis then compares the impact of the U.S.–China trade war to that of other exogenous shocks including the GFC and the COVID-19 pandemic, on the conditional volatility of energy and agricultural commodity markets and their interlinkages. Specifically, our current study attempts to answer three research questions:

RQ1) How does volatility transmit within and between energy (i.e., crude oil) and agricultural (i.e., soybean, corn, wheat, rice, sugar, cotton, coffee, and cocoa) commodity markets according to the VAR-GARCH-BEKK model for the period between July 30, 2007, and April 1, 2021?

RQ2) How does the U.S.–China trade war impact the volatility of crude oil and agricultural commodity markets, and how does it impact covariance between these markets?

RQ3) How does the impact of the U.S.–China trade war differ from the impact of other exogenous shocks on the variances and covariances of the crude oil and agricultural commodities markets according to the volatility impulse response function approach?

Addressing this research gap is necessary because significant volatility transmissions between energy and agricultural commodity markets may reduce the accuracy of predicting commodity

prices. Therefore, understanding volatility transmission linkages and the spillover effects of exogenous shocks can help individuals and firms make better asset allocation and price stabilization decisions (Yip et al., 2020). Furthermore, better understanding the nature of spillover effect and the impacts of exogenous shock on the energy and agricultural commodity markets can help investors and policymakers identify the implications of volatility spillovers during uncertain moments.

This study provides some important implications. First, given that the complexity and opacity of today's economic events inhibit accurate prediction (Fridgen et al., 2015), suppliers can effectively optimize asset allocation strategies and risk management by understanding the impacts of economic events on the energy and agricultural commodity markets. Second, food price shocks precipitate social distress, food security, poverty and inequality (Nkang, Omonona, Yusuf, & Oni, 2012). This is especially true for agriculture-based economies, where people derive their livelihood from agricultural practices (Nakamya & Romstad, 2020); thus, this study can provide policymakers with insights into solutions that might prevent such situations. Third, studying the impact of different exogenous shocks on variance and covariance between energy and agricultural commodities and then comparing the impact of various exogenous shocks can identify useful information flow patterns, which is particularly critical for investors and portfolio managers (Mensi, Tiwari, et al., 2017; Pal & Mitra, 2019).

The rest of this paper is organized as follows: Section 2 outlines the econometric strategy, variables, and data source. Section 3 reports the results, and Section 4 discusses empirical findings. Section 5 concludes with key findings and discusses policy implications.

2. Research methodology

The VAR-GARCH-BEKK model is employed to examine the volatility spillovers between the energy and agricultural commodity markets. Furthermore, the volatility impulse response function analysis (VIRF) proposed by Hafner and Herwartz (2006) is computed from the estimated VAR-GARCH-BEKK model to investigate the impact of U.S.–China trade war and other exogenous shocks on the volatility of energy and agricultural commodities and their covariances.

2.1. The VAR-GARCH-BEKK model

The generalized autoregressive conditional heteroskedasticity (GARCH) model incorporates autoregressive structure in the heteroscedastic variance, and multivariate GARCH models have been designed to model the conditional covariance matrix of multiple return series (Hafner & Herwartz, 2006), including the multivariate VAR-GARCH-BEKK model. However, as price returns usually respond in a nonlinear fashion to market changes, failure to account for asymmetry in the volatility models may cause model misspecification; thus, it is essential to allow asymmetry in volatility transmission (Abdelradi & Serra, 2015; Hasanov et al., 2016). Accordingly, our empirical analysis is based on an asymmetric specification of BEKK variance-covariance representation employed by Grier et al. (2004).

The mean model takes the following VAR form:

$$\mathbf{r}_t = c + \sum_{i=1}^p \mathbf{\Omega}_i \mathbf{r}_{t-i} + \mathbf{u}_t \quad (1a)$$

where $\mathbf{r}_t$ is vectors of returns, $\mathbf{\Omega}_i$ is coefficient matrix of lagged $\mathbf{r}_t$ process, and $\mathbf{u}_t$ is the vector of errors.

A variance-covariance of the BEKK specification is represented as

$$H_t = C'C + A'u_{t-1}u_{t-1}'A + B'H_{t-1}B + D'\xi_{t-1}\xi_{t-1}'D \quad (1b)$$

where

$$C = \begin{bmatrix} c_{11} & 0 \\ c_{21} & c_{22} \end{bmatrix}; A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}; B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}; D = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix}; \xi_{t-1} = \begin{bmatrix} \xi_{r_1 t} \\ \xi_{r_2 t} \end{bmatrix}$$

The matrix H_t is the conditional variance-covariance matrix, C is a lower triangular matrix, and A , B , and D are 2×2 parameter matrixes. This parameterization considers lagged conditional variances and covariances, H_{t-1} , as well as past values of $u_{t-1}u_{t-1}'$ and $\xi_{t-1}\xi_{t-1}'$, in joint estimations of contemporaneous volatilities of energy and agricultural commodities.

The diagonal elements of matrix A (a_{11}, a_{22}) represent the ARCH effect, which is the effect of lagged shocks. The off-diagonal elements of matrix A (a_{12}, a_{21}) represent the crossover-spillover effect. Besides, the diagonal elements of matrix B (b_{11}, b_{22}) represent the GARCH effect, which is the lagged volatility effect. The off-diagonal elements of matrix B (b_{12}, b_{21}) represent the crossover-volatility spillover effect. Hence, the off-diagonal elements of matrixes A and B represent the shock spillover effects and volatility spillover effects, respectively. Matrix D ($d_{11}, d_{22}, d_{12}, d_{21}$), $\xi_{t-1}\xi_{t-1}'$ in this case, takes into account potential asymmetric responses. The asymmetric impact on conditional variance indicates the influences of positive returns on volatility are different from the negative returns of a similar magnitude. Given the nature of the data, we assume generalized error distributed (GED) innovations, such that $\epsilon_t \sim GED(\nu)$ with ν being the degrees of freedom. The VAR(p)-GARCH-BEKK model is estimated by employing the maximum likelihood method, which finds the values of the parameters by maximizing the probability of obtaining the observed return data. The optimal lags for the model will be determined based on four information criteria, namely Akaike Information criterion (AIC), Schwarz's Bayesian criterion (SBC), Hannan–Quinn criterion (HQC), and final prediction error (FPE).

2.2. The volatility impulse response function (VIRF) analysis

The VIRF is a tool that traces information on the impact of independent shocks on volatility through time (Hafner & Herwartz, 2006). It captures how a shock in one market influences the dynamic adjustment of volatility in another market, and the persistence of these transmission effects. The VIRF focuses on shocks' impact on conditional variance rather than the conditional mean (Eraslan & Ali, 2018). Thus, we select VIRF to examine how conditional volatilities of energy commodity and agricultural commodities change in response to an exogenous shock, such as the U.S.–China trade war.

The VIRF tool has various distinctive features (Hafner & Herwartz, 2006): (i) it is a symmetric function of the shock with $\vartheta_t(\nu_0) = \vartheta_t(-\nu_0)$; (ii) it is not a homogeneous function of the shocks; (iii) it depends on the history through the initial volatility state at the time when the shock hits the system; and (iv) the decay or persistence of shocks is measured by the moving average matrixes $\Phi_t = (R + F)^{t-1}R$, which is analogous to the traditional analysis.

The VIRF is defined as the difference between the expected volatilities conditional on the initial shock and the available information set and the information set only:

$$\vartheta_t = E[\text{vech}(H_t)|v_0, F_{t-1}] - E[\text{vech}(H_t)|F_{t-1}] \quad (2)$$

where $\vartheta_t = [\vartheta_{o,t}, \vartheta_{os,t}, \vartheta_{s,t}]'$ is a three-dimensional vector containing the responses of the conditional variances of energy commodity and agricultural commodity market returns on its first and third elements, respectively, while the second element is the response function of the conditional covariance between the two markets returns. Moreover, v_0 and F_{t-1} denote the volatility shock and the information set available up to the period -1 , respectively, where v_0 is specified as $v_0 = H_0^{-\frac{1}{2}}\epsilon_0$.

Consequently, the initial response at time $t = 1$, which is the impact of a shock at time $t = 0$, is obtained as

$$\vartheta_1 = A^* \{\text{vech}(\epsilon_0\epsilon_0') - \text{vech}(H_0)\} \quad (3)$$

whereas the response function for any $t \geq 2$ is calculated as

$$\vartheta_t = (A^* + G^*)\vartheta_{t-1} \quad (4)$$

with A^* and G^* being $N^* \times N^*$ parameter matrixes expressed in the vech representation of the GARCH-BEKK model.

2.3 Exogenous shocks

Exogenous shocks refer to the random and unpredictable events that have a widespread impact on the economy, which are driven by things outside the scope of economic models (Fridgen et al., 2015). Markets are interconnected within economies and thus large shocks in any economic sector can have a far-reaching (positive or negative) macroeconomic impact.

Kilian and Park (2009) categorize the structural shocks of the oil market into (i) oil supply shocks, (ii) aggregate demand shocks, and (iii) oil-specific demand shocks ('precautionary demand shock'). Oil supply shocks refer to shocks to the global crude oil supply. Aggregate demand shocks refer to shocks to demand of all industrial commodities (including crude oil), driven by real global economic activity. Precautionary demand shocks refer to shifts in the precautionary demand for

crude oil due to high levels of uncertainty about future oil supply shortfalls. Following Eraslan and Ali (2018), who list historical exogenous shocks dominated by each type of oil price shocks depending on its origin, we include exogenous shocks between 2008 and 2017 in this analysis.

Furthermore, to account for the possible effects of unexpected exogenous shocks, our analysis also includes additional aggregate demand-side shocks. The additional aggregate demand shocks included in this analysis are the 2010 Venezuela crisis, the 2011 Japanese Tsunami, the Greek debt crisis that threatened the European Union, Brexit, and the COVID-19 pandemic and subsequent 2020 recession (Su, 2014; Amadeo, 2020a; Clayton, 2016). Table 1 lists the selected exogenous shocks.

Table 1: Selected exogenous shocks

Timeline	Event	Shock	Reference
15.9.2008	The global financial crisis of 2007–2008	Aggregate demand-side	Eraslan and Ali (2018)
4.6.2010	Venezuela Crisis 2010	Aggregate demand-side	Su (2014)
10.1.2011	Libya's unrest and the subsequent NATO intervention, and Saudi Arabia's increase in its oil production	Supply-side	Eraslan and Ali (2018)
11.3.2011	Japan's Tsunami 2011	Aggregate demand-side	Amadeo (2021); Clayton (2016)
8.8.2011	The downgrade of the U.S. debt status in August 2011	Aggregate demand-side	Eraslan and Ali (2018)
30.4.2012	The European sovereign debt crisis	Aggregate demand-side	Eraslan and Ali (2018)
14.7.2014	Robust global production exceeded	Aggregate demand-side	Eraslan and Ali (2018)
30.6.2015	Greek Debt Crisis Threatens European Union	Aggregate demand-side	Amadeo (2021); Clayton (2016)
23.6.2016	Brexit	Aggregate demand-side	Amadeo (2021); Clayton (2016)
2.12.2016	OPEC and non-OPEC producers reached their first deal since 2001 to curtail oil output jointly (Nearest structural break in crude oil)	Supply-side	Eraslan and Ali (2018)
7.4.2017	The U.S. missile strike of Syria's Shayrat Airbase	Precautionary demand	Eraslan and Ali (2018)
22.5.2017	OPEC and non-OPEC members agree to extend production cuts for nine months	Supply-side	Eraslan and Ali (2018)
6.7.2018	U.S.–China trade war	Aggregate demand-side	(Bakst & Beaumont-Smith, 2018; Bloomberg News, 2018)
11.3.2020	COVID-19 Pandemic and 2020 Recession	Aggregate demand-side	Amadeo (2021)

2.4. Data and summary statistics

Brent crude oil price is selected as the proxy for the energy market, while soybean, corn, wheat, rice, sugar, cotton, coffee and cocoa prices are used to represent the agricultural commodity

markets.⁷ Notably, futures settlement prices are used because they provide a better analysis of the commodities prices due to higher trading volume and liquidity (Cuny, 1993). They are collected from DataStream at the daily frequency, covering the period from July 30, 2007, to April 1, 2021, according to their data availability. More details on these futures contracts are given in Table 2. Fig. 2 plots the time evolution of the price levels and log returns of the nine futures contracts under study. We notice significant drops in the price of crude oil around the 2008 global financial crisis and the COVID-19 outbreak. A very similar pattern, but with a different magnitude in the price and return movement, is observed for soybeans and corn. They rebounded more sharply following the peak of the pandemic, after experiencing some flat to a slow decline during 2018–2019.

Table 2: Data source details.

Name	Series name	Unit	Source
Brent crude oil	NYM-BRENT CRUDE OIL LST DAY CONT. – SETT. PRICE – U\$/BL	U\$/Barrels	New York Mercantile Exchange
Soybean	CBT-SOYBEAN COMP. CONT. – SETT. PRICE	UC/Bushels	Chicago Board of Trade
Corn	CBT-CORN COMP. CONTINUOUS – SETT. PRICE – UC/BL	UC/BL	Chicago Board of Trade
Wheat	CBT-WHEAT COMPOSITE FUTURES CONT. – SETT. PRICE – UC/HP	UC/HP	Chicago Board of Trade
Rice	CBT-ROUGH RICE COMP FUTURES CONT. – SETT. PRICE – UC/HP	UC/HP	Chicago Board of Trade
Sugar	CSCE-SUGAR #11 CONTINUOUS – SETT. PRICE – UC/LB	UC/LB	ICE Futures U.S.
Cotton	CSCE-COTTON #2 CONTINUOUS – SETT. PRICE – UC/LB	UC/LB	ICE Futures U.S.
Coffee	CSCE-COFFEE ‘C’ CONTINUOUS – SETT. PRICE – UC/LB	UC/LB	ICE Futures U.S.
Cocoa	CSCE-COCOA CONTINUOUS – SETT. PRICE – U\$/TE	U\$/TE	ICE Futures U.S.

The price series (P_t) are converted into log returns series (R_t) as follows

⁷ Using agricultural commodities for this purpose builds on various previous empirical studies (Ahmadi et al., 2016; Bakas & Triantafyllou, 2018; Balcilar et al., 2018; Eraslan & Ali, 2018; Hasanov et al., 2016) to arrive at six agricultural commodities categories: cereal grains, oilseeds, meat, dairy, soft commodities, and miscellaneous agricultural commodities. From these categories, on the basis of high liquidity and trading volume, the focus is narrowed to futures data for oilseeds (soybean and cotton), cereal grains (corn, wheat, and rice) and soft commodities (sugar, cocoa, and coffee), an approach designed to comprehensively examine volatility transmission between energy and agricultural commodities (Amadeo, 2020).

$$R_{it} = 100 \times \ln \left(\frac{P_{it}}{P_{it-1}} \right) \quad (5)$$

where $i = 1$ through 9, denoting the nine commodities under study.

Table 3: Descriptive statistics of log returns

	Crude oil	Soybean	Corn	Wheat	Rice	Sugar	Cotton	Coffee	Cocoa
Mean	-0.0043	0.0149	0.0154	-0.0012	0.0069	0.0108	0.0062	0.0021	0.0060
Std. Dev.	2.3224	1.5029	1.7970	2.0287	1.5554	2.0981	1.8058	1.9601	1.8216
Min.	-27.9762	-14.0831	-24.5286	-9.9728	-16.1242	-12.3658	-27.1398	-11.2541	-12.5137
Max.	19.0774	6.4760	8.6618	12.9293	9.8022	13.0620	9.5310	11.7892	11.4913
Skewness	-0.7937	-0.8176	-0.7220	0.1829	-0.5777	-0.0022	-1.1447	0.0939	-0.2798
Kurtosis	19.4434	9.8299	14.9832	5.4811	11.5020	6.2811	19.9377	4.8063	6.3438
ADF	-57.2400 ***	-52.7890 ***	-53.7680 ***	-54.5220 ***	-48.7030 ***	-54.1580 ***	-51.8410 ***	-57.6170 ***	-55.0320 ***
PP	-57.0790 ***	-52.7890 ***	-53.8960 ***	-54.5730 ***	-48.7090 ***	-54.1830 ***	-51.8260 ***	-57.5640 ***	-55.0700 ***

Notes: The sample period is period July 30, 2007, to April 1, 2021, yielding 3,468 daily return observations per series. ADF denotes the Augmented Dickey-Fuller unit root test; PP denotes the Phillips–Perron unit root test. *** Indicates statistical significance at 1%.

Table 3 presents the descriptive statistics for the log return series for the full sample (July 30, 2007, to April 1, 2021). All average returns are positive, except crude oil and wheat. Crude oil, wheat, and sugar exhibit the highest volatility. The kurtosis for all return series is larger than 3, indicating that all series are leptokurtic. The return distribution of crude oil, soybeans, corn, rice, sugar, cotton, and cocoa is negatively skewed. Given that GARCH models require stationary series, we apply the Augmented Dickey and Fuller (ADF) and Phillips and Perron (PP) tests and ensure at the 1% significance level that all return series follow a stationary process.

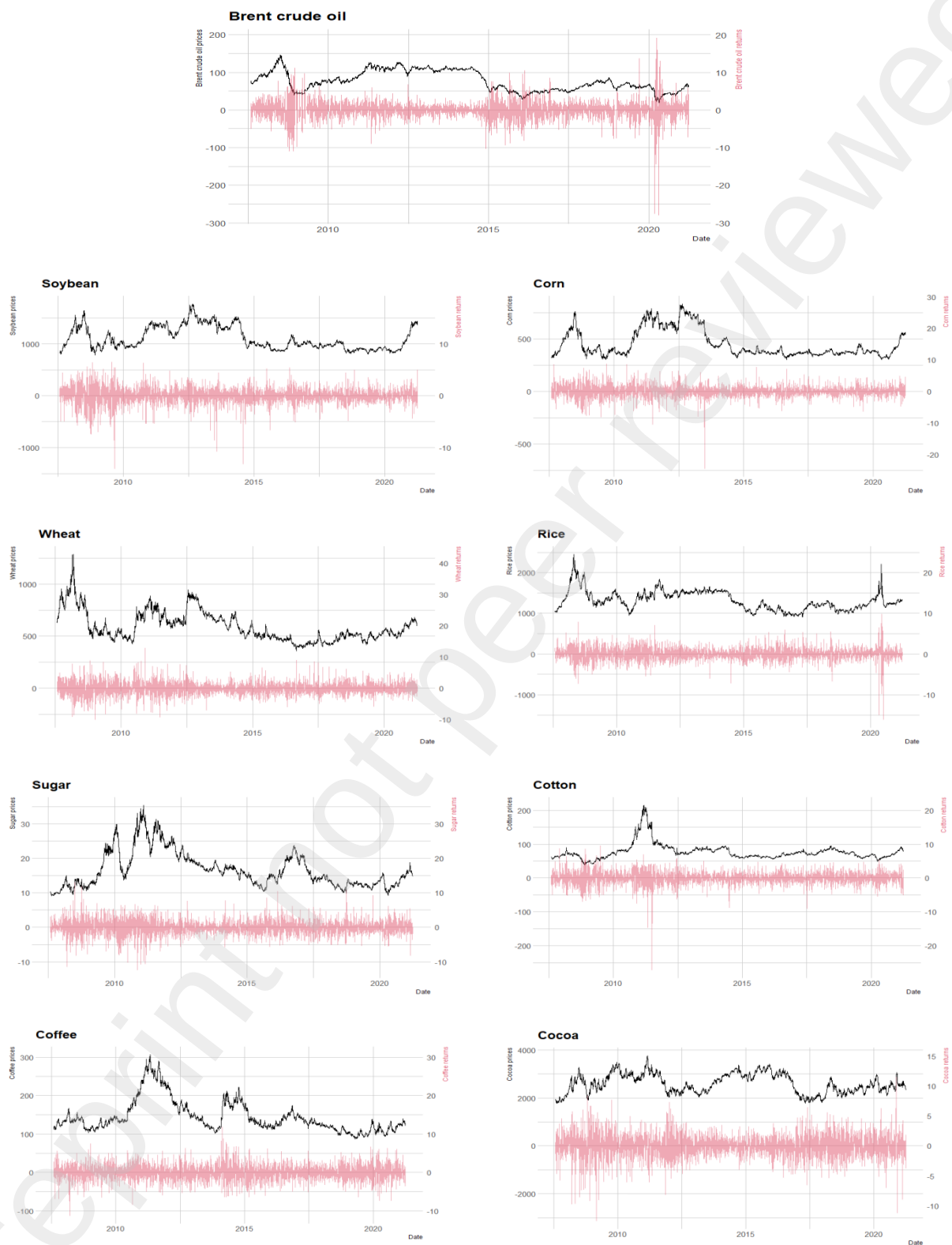


Fig. 2. The prices of crude oil and agricultural commodities and their log- returns for the full sample period (August 1, 2007 to April 1, 2021).

3. Empirical results

Following the estimation of the VAR-GARCH-BEKK model, we present in Fig. 3 the conditional volatility series for each of the nine commodities under study, where the highlighted cluster reflects the period of the U.S.–China trade war (July 6, 2018 to April 1, 2021) (Wong & Koty, 2019). Volatility clustering tends to occur simultaneously between the crude oil and some of the agricultural commodities, suggesting potential volatility spillover effects.

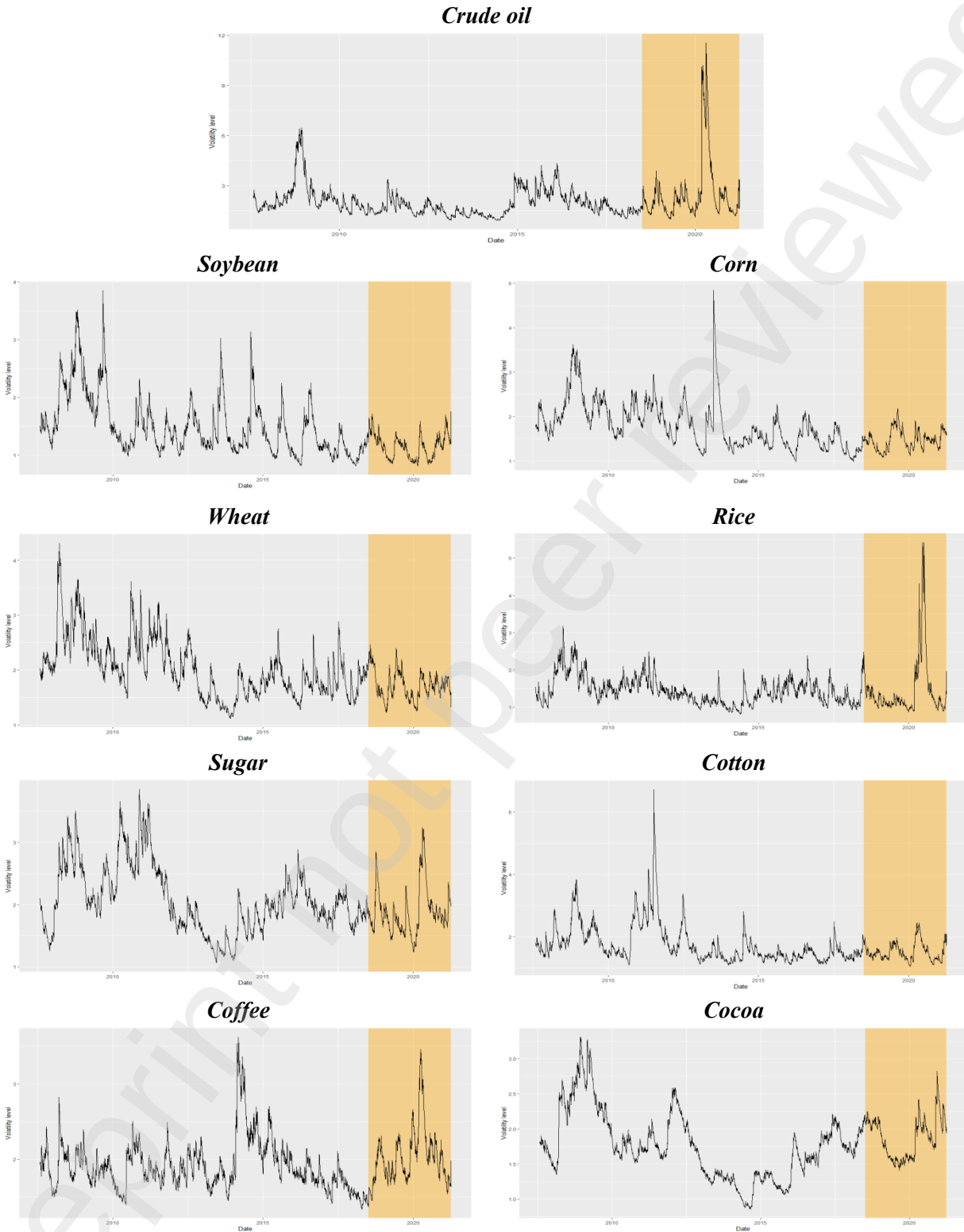


Fig. 3. *Volatilities of crude oil and agricultural commodity returns for the sample period (August 1, 2007 to April 1, 2021). The highlighted period is the U.S.–China trade war period (July 6, 2018 to April 1, 2021).*

3.1. The volatility transmissions between crude oil and agricultural commodity markets

Table 4 reports the estimated⁸ coefficients of matrixes C, A, B and D from Equation (1b). The analysis employs four information criteria—AIC, SBC, HQC and FPE—to select the optimal lags for the model. The maximum lag order is set to 5 to ensure a sufficient degree of freedom and avoid numerical convergence problems. Having established that the optimal model is the VAR(1)-GARCH-BEKK model, it is used for model specification and estimation of volatility transmission and VIRF analysis.

The element a_{ij} in matrix A represents the effect of shocks arising from commodity market i on market j . The subscript $i = 1$ corresponds to crude oil commodity, and $j = 2$ corresponds to an agricultural commodity (soybeans, corn, wheat, rice, sugar, cotton, coffee, or cocoa). The diagonal elements in matrix A (a_{11}, a_{22}) evaluate the effect of past shocks (GARCH effects), and the off-diagonal elements of matrix A (a_{12}, a_{21}) represent the spillover effect, which indicates the cross-market influences of shocks. If the parameter estimates for a_{1j} are significant, then the volatility of the agricultural commodity is affected by shocks from the crude oil market. Table 4 shows that the results indicate that the parameter estimates for a_{1j} for wheat and coffee are not significant. However, the volatilities of the soybean, corn, rice, sugar, cotton, and cocoa markets are affected by shocks from the crude oil market. Furthermore, the coefficients $a_{1,1}$ and $a_{2,2}$ are statistically significant at the 1% significance level, indicating that the volatilities of all of the commodities are impacted by shocks.

⁸ The model estimations were conducted using RATS 10.0.

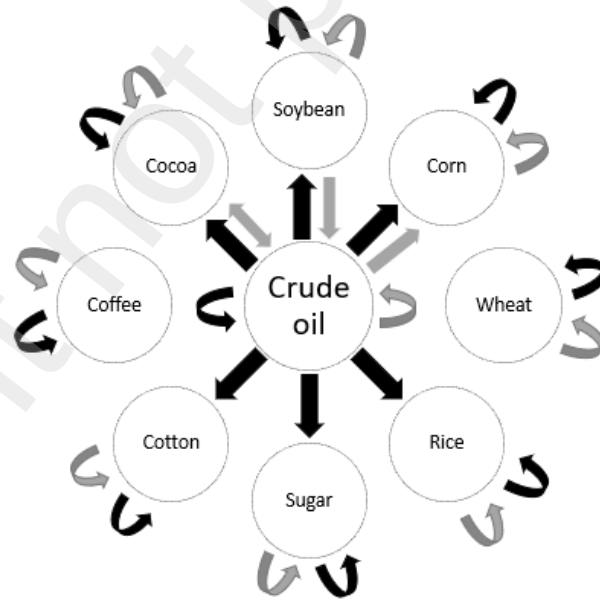
Table 1: Estimates for parameter matrixes of conditional asymmetric BEKK variance-covariances and diagnostic and specification test results for the estimated VAR-GARCH-BEKK model

Variable	Soybean		Corn		Wheat		Rice		Sugar		Cotton		Coffee		Cocoa	
C(1,1)	0.1748***		0.1798***		0.1957***		0.1797***		0.1835***		0.1782***		0.2012***		0.1892***	
C(2,1)	-0.0450*		-0.035		-0.1265***		-0.0099		-0.0307		-0.0134		0.0396		-0.0173	
C(2,2)	0.1018***		0.1193***		0.0948		0.1523***		0.0976***		0.1666***		0.1830***		0.0549**	
A(1,1)	0.1408***		0.1328***		0.1340***		0.1321***		0.1516***		0.1299***		0.1533***		0.1553***	
A(1,2)	0.0105***		0.0138**		0.0126		0.0427***		-0.0240**		0.0169**		-0.0075		0.0180**	
A(2,1)	0.0028		0.0203		-0.0061		0.0394		0.0044		-0.0135		0.0131		-0.0195	
A(2,2)	0.1650***		0.1637***		0.1636***		0.1899***		0.1289***		0.1729***		0.1459***		0.1163***	
B(1,1)	0.9588***		0.9599***		0.9551***		0.9576***		0.9559***		0.9580***		0.9557***		0.9531***	
B(1,2)	-0.0018		-0.0053***		-0.0013		-0.0052		-0.002		0.0007		-0.0017		-0.0063**	
B(2,1)	0.0132**		0.0035		0.0148*		0.0007		0.0087		0.0081		-0.0025		0.0104***	
B(2,2)	0.9738***		0.9754***		0.9775***		0.9623***		0.9834***		0.9692***		0.9784***		0.9902***	
D(1,1)	0.2150***		0.2216***		0.2473***		0.2315***		0.2191***		0.2322***		0.2337***		0.2213***	
D(1,2)	0.0083		0.0218**		0.0061		-0.0049		0.0774***		-0.0065		0.0479***		0.0091	
D(2,1)	0.0109		0.0111		0.0248		0.0614**		0.0289		0.0467*		0.0094		0.0627***	
D(2,2)	-0.0012		0.0121		-0.0075		-0.0574		-0.0989***		0.0147		-0.0288		0.0078	
Univariate tests	crude oil	soybean	crude oil	corn	crude oil	wheat	crude oil	rice	crude oil	sugar	crude oil	cotton	crude oil	coffee	crude oil	cocoa
Ljung–Box Q (12)	9.0196	7.1132	9.0471	13.5370	10.3602	8.3577	10.2814	14.4247	9.5437	7.8634	9.1431	4.2549	9.1126	11.6082	8.4409	11.3469
Ljung–Box Q (24)	15.4970	19.2371	15.7161	32.8086	16.1946	23.3813	16.2382	25.1300	15.4018	19.6368	15.5384	15.8656	15.5659	41.1816**	14.8841	26.8167
McLeod–Li (12)	18.7796*	6.9221	18.4614	1.5810	18.9682*	10.2770	19.3923*	9.8944	17.7484	15.5749	22.8375**	3.1608	18.5823*	14.3247	18.1802	25.8112**
McLeod–Li (24)	28.4954	71.4072***	27.8040	2.6189	27.7389	17.3955	29.1349	25.1130	27.4047	28.2117	33.0380	8.0268	27.9322	30.0303	27.3744	33.9416*
ARCH LM (5)	2.6430**	0.8530	2.6150**	0.2200	2.8670**	0.6390	2.8890**	1.3850	2.4980**	2.3630**	3.6030***	0.4550	2.6160**	1.9820*	2.5910**	0.4290

Notes: This table presents the estimates of conditional asymmetric BEKK variance-covariances and diagnostic and specification test results for the estimated VAR-GARCH-BEKK model. Matrix A represents the effect of shocks arising from commodity market i on market j ; matrix B represents the effect of past volatility on current conditional variances; matrix D represents the hypothesis of an asymmetric covariance process. *** Indicates statistical significance at 1%; ** Indicates statistical significance at 5%; * Indicates statistical significance at 10%.

The element b_{ij} in matrix B represents the effect of past volatility on current conditional variances. The diagonal elements in matrix B (b_{11}, b_{22}) capture the GARCH effect of a commodity's own lagged volatility on its contemporaneous volatility. The off-diagonal elements in matrix B (b_{12}, b_{21}) indicate the volatility of cross-market spillover impacts. That is, the volatility spillover from market i to market j for element b_{ij} in matrix B. Table 4 indicate significant parameter estimates for b_{1j} for corn and cocoa, suggesting that the past volatility of crude oil returns significantly impacts the current volatility of corn and cocoa commodities returns. Additionally, the results suggest significant parameters estimated for b_{21} for soybean and cocoa, which indicates that the past volatility of soybean and cocoa returns significantly impacts the current volatility of crude oil commodities returns. Nonetheless, all of the diagonal elements in matrix B (b_{11}, b_{22}) are statistically significant at the 1% level, implying that the conditional volatilities are associated with past conditional volatilities in all markets. Moreover, the magnitude of the diagonal parameter estimates shows that the conditional volatilities are highly related to past conditional volatilities because they are close to unity.

Fig. 4: The volatility linkages between crude oil and agricultural commodity markets



Notes: The black arrows indicate that there is a significant effect of shocks arising from commodity market i on market j . The gray arrows indicate that there is a significant effect of past volatility of commodity market i on the current conditional variances of market j .

Fig. 4 presents the summary of the results of the volatility transmission between crude oil and agricultural commodities. The directed arrow indicates the direction of the volatility spillovers between the commodities which is in accordance with Table 4. The black arrows indicate significant shock transmission from market i to market j , and the gray arrows indicate significant volatility spillover from market i to market j .

Element d_{ij} in matrix D represents the hypothesis of an asymmetric covariance process, which is based on the research of Grier et al. (2004). The significance of parameter matrix D suggests that negative price shocks lead to more price volatility than positive price shocks of the same magnitude. The results indicate that crude oil (d_{11}) and sugar (d_{22}) feature variance asymmetry. Additionally, the results from Table 4 reveal statistically significant cross-price asymmetric responses for d_{12} in the model estimates for crude oil–corn, crude oil–sugar, and crude oil–coffee and for d_{21} in the model estimates for rice–crude oil, cotton–crude oil and cocoa–crude oil. Negative price shocks in market i lead to more price volatility in market j than positive shocks.

Table 4 reports the diagnostic and specification test results for the estimated VAR-GARCH-BEKK model. The Ljung–Box Q statistic is a univariate version of the serial correlation test for standardized residuals of lags 12 and 24. The results show that the standardized errors in the equations are not serially dependent up to lag 12 and lag 24 for any agricultural commodity case at the 1% significance level. The McLeod–Li test determines serial dependence on the univariate squared standardized residuals, with results suggesting that no agricultural commodities return series are serially dependent up to lag 12 and lag 24 at the 1% level of significance except for soybeans at lag 24 and cocoa at lag 12. The ARCH LM (5) test statistic inspects the remaining heteroskedasticity up to lag 5. No agricultural commodity presents a significant result at the 1% significance level. The McLeod–Li test and ARCH LM test results do not demonstrate significant results at the 1% level, indicating that the considered series do not include violations, and supporting the specifications for the variance-covariance process adopted in this study.

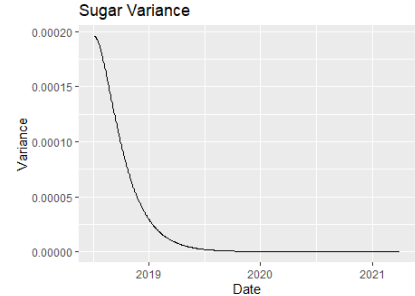
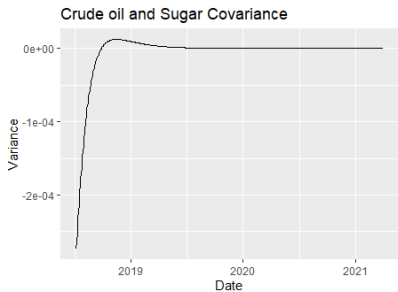
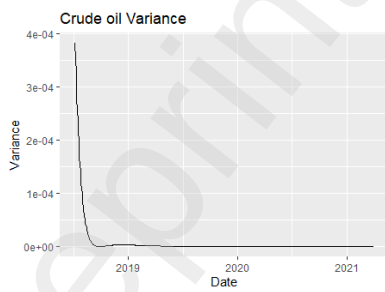
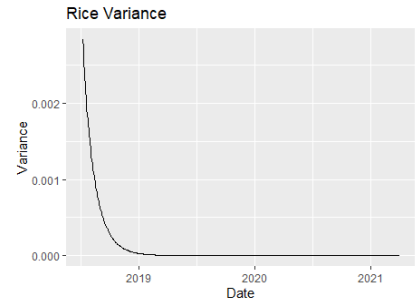
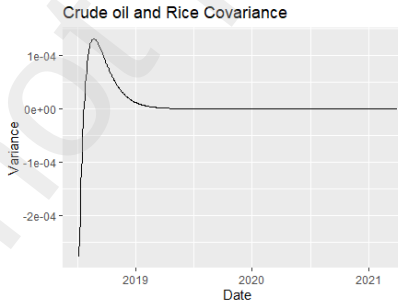
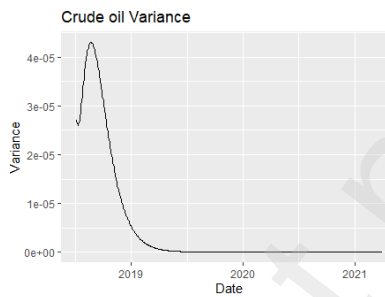
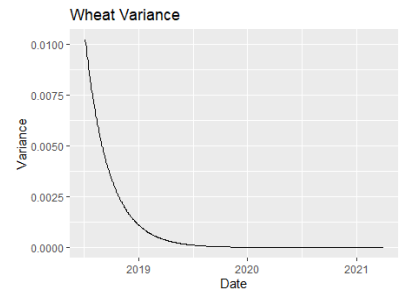
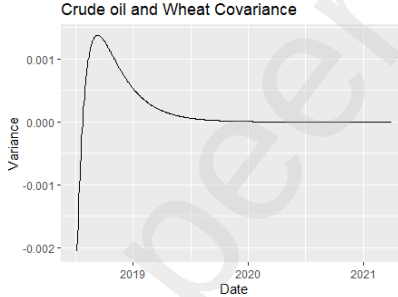
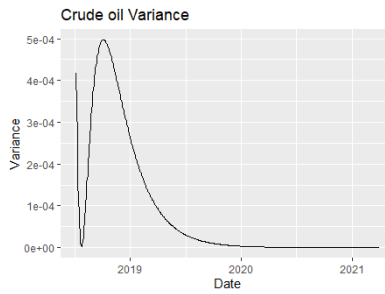
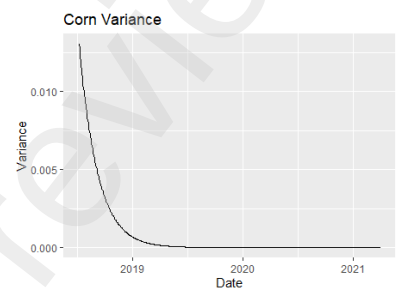
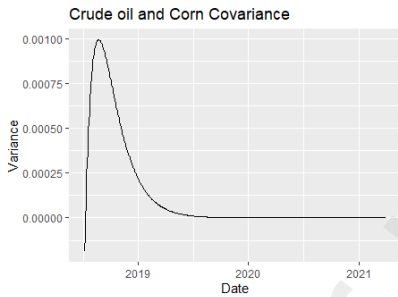
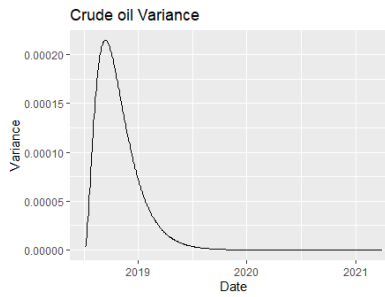
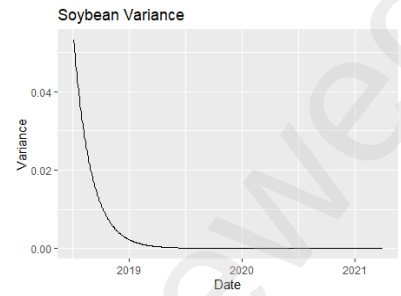
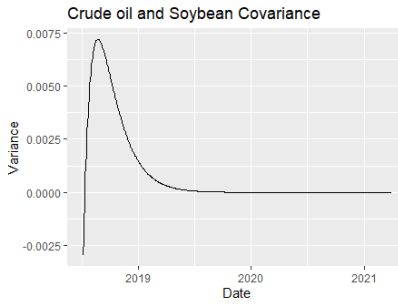
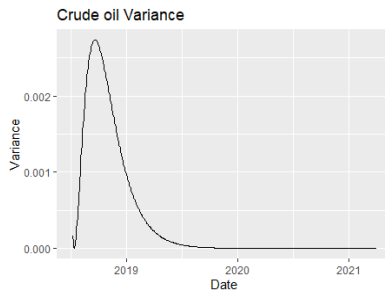
3.2. The impact of the U.S.–China trade war on the variance and covariance of crude oil and agricultural commodities

In this section, we focus on Fig. 5 and Table 5, which represent the impulse response function for the covariance between crude oil and agricultural commodity returns and contain the variances of changes in individual crude oil and agricultural commodities prices. Although some events leading up to the U.S.–China trade war spanned several months, the United States' first China-specific tariffs were imposed on July 6, 2018. Hence, this date is used as the base point for the analysis.

Table 5: Statistics for the volatility impulse response functions for the U.S.–China trade war starting on July 6, 2018 (In reference to Fig. 5).

	Covariance				Agricultural Variance			
	Initial point	Terminal point	Maximum absolute value	Vanish completely	Initial impact	Terminal point	Maximum absolute value	Vanish completely
Soybean	-0.2970%	0.0000%	0.7188%	YES	5.3227%	0.0000%	5.3227%	YES
Corn	-0.0188%	0.0000%	0.0995%	YES	1.3041%	0.0000%	1.3041%	YES
Wheat	-0.2063%	0.0000%	0.2063%	YES	1.0206%	0.0000%	1.0206%	YES
Rice	-0.0277%	0.0000%	0.0277%	YES	0.2841%	0.0000%	0.2841%	YES
Sugar	-0.0274%	0.0000%	0.0274%	YES	0.0196%	0.0000%	0.0196%	YES
Cotton	-0.5800%	0.0000%	0.5800%	YES	3.5523%	0.0000%	3.5523%	YES
Coffee	-0.0210%	0.0000%	0.0345%	YES	4.0985%	0.0000%	4.0985%	YES
Cocoa	0.0411%	0.0001%	0.0411%	No	0.0766%	0.0002%	0.0766%	No

Notes: This table presents the impulse response function for the covariance between crude oil and agricultural commodity returns and contains the variances of changes in individual agricultural commodities prices.



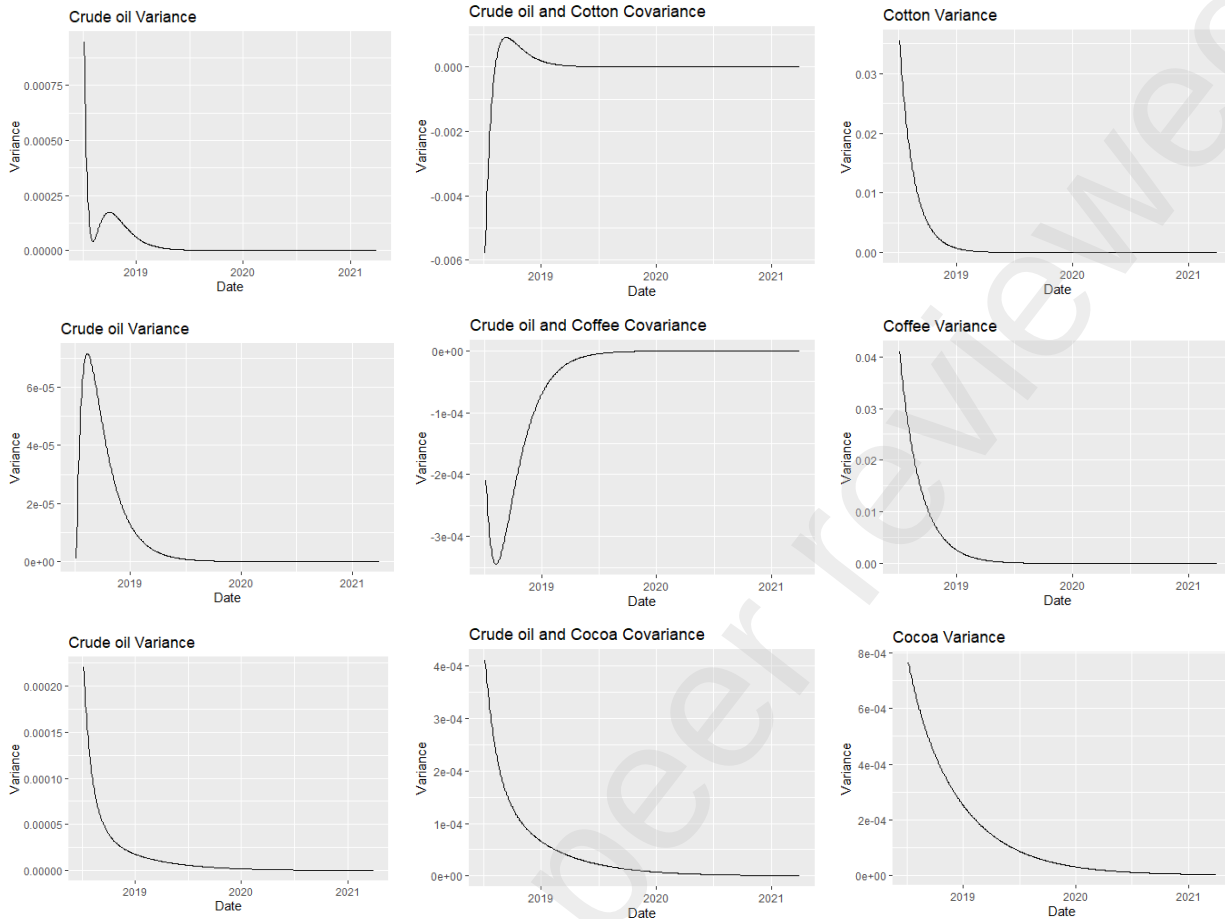


Fig. 5. Volatility impulse response functions for the U.S.–China trade war starting on July 6, 2018. First column: variance of crude oil price returns; second column: covariance between crude oil and each agricultural commodity; third column: variance of each agricultural commodity price returns.

As Fig. 5 shows, we notice some volatility increases result from this shock on all conditional variances, indicating that the U.S.–China trade war significantly impacts crude oil and agricultural commodity markets. However, the size of the effects is not uniform across all agricultural commodities, with results revealing relatively large volatility increases resulting from this shock on the conditional variances of soybeans, coffee, and cotton (approximately 5.32%, 4.10% and 3.55%, respectively). Furthermore, small volatility increases for corn and wheat deriving from this shock can be observed, with conditional variances increasing by about 1.30% and 1.02%, respectively. However, the sizes of the increased conditional volatilities for rice, cocoa, and sugar price returns are much smaller compared to other agricultural commodities, with an initial positive impact of the shock on these three commodities of less than 0.3%. Thus, soybeans, coffee, and

cotton exhibit greater responsiveness to shocks than other commodities. Furthermore, the impact of the U.S.–China trade war on the volatilities of prices of agricultural commodities appears to vanish completely towards the end of the first quarter of 2021 for all commodities except cocoa. Meanwhile, crude oil variances are responsive to the historical shock, despite the magnitude of the impact being relatively small.

The results also reveal the impact of the shock on the covariance between crude oil and each agricultural commodity. The largest impact is observed for the covariance between crude oil and soybeans, initially recorded at about -0.30% , and increasing over time to a positive value in November 2018 before reaching its peak of 0.72% in October 2018. It begins to exhibit a gradual decrease by the beginning of October 2018, with the impact of the U.S.–China trade war vanishing completely at the end of the first quarter of 2021. Elsewhere, the crude oil–corn covariance is initially observed at approximately -0.02% and increases to reach a peak of 0.10% before decreasing.

The crude oil–wheat covariances and crude oil–cotton covariances exhibit similar behavior, recording initial impacts in the form of negative values (-0.21% and -0.58% , respectively) before increasing over time to peak in September 2018 at about 0.14% and 0.09% , respectively. Afterwards, they trend backwards. Additionally, crude oil–rice and crude oil–sugar covariances are initially observed at about -0.03% . Both trend upwards until the end of August 2018 for crude oil–rice covariance and November 2018 for crude oil–sugar covariance. This is followed by downward trends for both, with the impacts of shock on the covariances of crude oil–rice and crude oil–sugar vanishing at the end of 2020 and the end of March 2021, respectively. For crude oil–coffee covariance, the initial observation is -0.02% , which increases to -0.03% and vanishes gradually after that. For crude oil–cocoa covariance, the initial observation is 0.04% , which begins to decline immediately following this peak point. Thus, the shock has the largest impact on crude oil–soybean (0.72%), crude oil–cotton (0.58%) and crude oil–wheat (0.21%) covariances, with the shock on all covariances (except for cocoa), ultimately appearing to vanish completely.

Our results demonstrate that the U.S.–China trade war has caused more volatility in soybeans, coffee, and cotton than in corn, wheat, rice, cocoa and sugar price returns. They also indicate that the U.S.–China trade war has considerably impacted the volatilities of energy and agricultural

commodities price returns and their covariance. However, the size and dynamic of the impacts of the shock are largely commodity-specific.

3.3. The impact of different exogenous shocks on the variance and covariance between crude oil and agricultural commodities markets

The previous section revealed that the U.S.–China trade war's impact on the price changes of energy and agricultural commodities market is commodity-specific. This section aims to respond to the third research question by comparing these findings with the impact of other exogenous shocks on the volatilities of energy and agricultural commodity markets and the volatility linkage between them. As previously discussed, oil price shocks and other economic shocks, including resource shortage, natural disasters and economic crises, are considered exogenous shocks in the context of this paper.

Table 6 shows the impact of the selected exogenous shocks on agricultural commodity markets. The effects of these exogenous shocks on most agricultural commodity prices are generally only marginally significant, with these results showing that the impact on agricultural commodities is not consistent across different exogenous shocks and is commodity-specific. As detailed, for three out of eight commodities, the volatility changes observed in response to the U.S.–China trade war are marginally significant. While the magnitudes of the responses of soybeans, cotton, coffee, followed by corn and wheat to the U.S.–China trade war are greater than the magnitudes of their responses to other exogenous shocks, the price volatility of rice, cocoa, and sugar in response to the U.S.–China trade war is smaller than the volatility following other shocks, indicating that the U.S.–China trade war has had a weaker impact on these agricultural markets. These results indicate that the impact of the trade war has, in general, a considerable impact on agricultural commodities compared to other exogenous shocks.

Table 6 shows the impact of shocks on the price volatility of soybean commodities between 0.00% and 7.81%. Only the Greek debt crisis (7.81%) records a greater impact on soybeans than the U.S.–China trade war (5.32%). The impacts of other shocks on soybean price volatility are observed between 0.00% and 1.58%. Similarly, only one exogenous shock, the global financial crisis (5.15%), has a more considerable impact on cotton than the U.S.–China trade war (3.55%). The impacts of other exogenous shocks on the price volatility of cotton are smaller, ranging

between 0.00% and 3.10%. The impact of the U.S.–China trade war (4.10%) on the price volatility of coffee is only lower than the impact of the Japanese tsunami (5.10%), with the impacts of other exogenous shocks on coffee's volatility are observed to be between 0.00% and 2.72%.

Four out of 12 exogenous shocks exhibit a larger impact on corn's price volatility than the impact of the U.S.–China trade war (1.30%), ranging between 1.34% and 16.78%, with the smaller impacts of the exogenous shocks on corn's price volatility ranging between 0.05% and 1.26%. Furthermore, five out of 12 exogenous shocks demonstrate a greater impact on wheat's price volatility than the U.S.–China trade war. The impact of the U.S.–China trade war on wheat is observed at 1.02%; the impacts of other exogenous shocks on wheat range between 0.01% and 9.53%. Only two exogenous shocks impact cocoa and rice price volatility less than the U.S.–China trade war, and all other exogenous shocks impact sugar price volatility more than the U.S.–China trade war (0.02%), with the impacts of other exogenous shocks on the price volatility of sugar ranging between 0.04% and 6.10%.

Table 7 shows the impact of exogenous shocks on the covariance between crude oil and agricultural commodities. The U.S.–China trade war's impacts on crude oil–soybean and crude oil–cotton covariances are larger than the corresponding impacts of other exogenous shocks. However, most other exogenous shocks have a greater impact on other covariances than the trade war. For example, six out of 12 exogenous shocks impact the covariance between crude oil and soybeans more than the U.S.–China trade war (0.72%), recording covariances ranging between 0.94% and 3.20%. The exogenous shocks have a smaller impact on the crude oil–soybean covariance range in magnitude between 0.03% to 0.48%. Meanwhile, six out of 12 exogenous shocks impact the crude oil–cotton covariance more than the trade war (0.58%), recording covariances ranging between 0.64% and 4.77%, with the smaller impacts ranging between 0.01% and 0.47%. Furthermore, only two exogenous shocks impact the crude oil–wheat covariance less than the trade war (0.21%), ranging in magnitude between 0.02% and 0.06%, with the exogenous shocks that impact the covariance more than the trade war ranging between 0.21% and 5.37%. Similarly, only two exogenous shocks impact the crude oil–coffee covariance less than the U.S.–China trade war (0.03%), recording magnitudes between 0.01% and 0.03%; the heavier-impact shocks demonstrate covariances between 0.11% and 3.82%.

Two exogenous shock demonstrates smaller impacts (0.08% and 0.05%) on the crude oil–corn covariance than the U.S.–China trade war (0.10%). The other shocks indicate covariances in the range between 0.11% and 6.29%. For crude oil–cocoa covariance, the two exogenous shocks' impacts are smaller (both 0.01%) than the impact of the U.S.–China trade war (0.04%), with the exogenous shocks with greater impacts recording covariances ranging between 0.11% and 3.56%. For the crude oil–rice covariance, the impact of the U.S.–China trade war (0.03%) demonstrates a larger impact than only the shock of OPEC and non-OPEC members agreeing to extend production cuts for nine months (0.02%), with the impacts of other exogenous shocks ranging between 0.08% to 8.10%. Finally, all exogenous shocks impact the crude oil–sugar covariance more than the U.S.–China trade war (0.03%), with the magnitudes ranging between 0.14% and 3.90%.

Table 6: The impact of exogenous shocks on the variance of agricultural commodities.

Timeline	Event	Shock	Soybean	Corn	Wheat	Rice	Sugar	Cotton	Coffee	Cocoa
15.9.2008	The global financial crisis of 2007–2008	Aggregate demand-side	1.4323%	0.1337%	0.1159%	0.7006%	0.0947%	5.1455%	0.0388%	0.3133%
4.6.2010	Crisis in Venezuela	Aggregate demand-side	1.5702%	2.6209%	0.7213%	1.5115%	3.4917%	1.0995%	0.2827%	3.4151%
10.1.2011	Libya's unrest and the subsequent NATO intervention and Saudi Arabia's increase of its oil production	Supply-side	0.3812%	2.2675%	1.5079%	0.7957%	6.0960%	2.4773%	0.0010%	0.1218%
11.3.2011	Japan's Tsunami	Aggregate demand-side	1.4120%	0.0538%	1.8082%	0.5229%	2.3226%	0.3571%	5.0992%	0.9515%
8.8.2011	The downgrade of the U.S. debt status in August 2011	Aggregate demand-side	1.5809%	1.3352%	5.6690%	1.2023%	0.4766%	0.0224%	1.0871%	0.2061%
30.4.2012	The European sovereign debt crisis	Aggregate demand-side	0.0856%	1.2565%	0.6852%	1.3692%	1.9494%	0.8103%	0.1532%	0.1852%
14.7.2014	Robust global production exceeded	Aggregate demand-side	0.0030%	0.2019%	1.2243%	1.8399%	0.0705%	0.0112%	0.6710%	0.0056%
30.6.2015	Greek Debt Crisis Threatens European Union	Aggregate demand-side	7.8181%	16.7787%	9.5328%	6.8798%	1.8711%	0.4930%	0.0203%	0.3469%
23.6.2016	Brexit	Aggregate demand-side	0.2946%	0.4636%	0.1863%	0.4739%	0.0445%	3.0951%	0.5017%	0.2057%
2.12.2016	OPEC and non-OPEC producers reached their first deal since 2001 to curtail oil output jointly (Nearest structural break in crude oil)	Supply-side	0.0643%	0.1276%	0.0350%	4.8063%	0.4671%	0.2997%	0.9300%	0.0847%
7.4.2017	The U.S. missile strike of Syria's Shayrat Airbase	Precautionary demand	0.1394%	0.0731%	0.1443%	0.4618%	0.6557%	0.0012%	0.0274%	0.0012%
22.5.2017	OPEC and non-OPEC members agree to extend production cuts for nine months	Supply-side	0.4953%	0.5592%	0.0094%	0.0129%	0.5109%	0.3690%	2.7241%	0.3171%
6.7.2018	U.S.–China Trade war	Aggregate demand-side	5.3227%	1.3041%	1.0206%	0.2841%	0.0196%	3.5523%	4.0985%	0.0766%
11.3.2020	COVID-19 Pandemic and 2020 Recession	Aggregate demand-side	0.0826%	0.3139%	0.9898%	0.0260%	0.5823%	0.0200%	0.6520%	0.3228%
No. of events that have larger impact than trade war			1	4	5	11	13	1	1	11

Notes: This table presents the impact of exogenous shocks on the variance of agricultural commodities. Bold numbers indicate that the event has a higher impact on a particular agricultural product compared to U.S.–China trade war.

Table 7: The impact of exogenous shocks on the covariance between crude oil and agricultural commodities.

Timeline	Event	Shock	Soybean	Corn	Wheat	Rice	Sugar	Cotton	Coffee	Cocoa
15.9.2008	The global financial crisis of 2007–2008	Aggregate demand-side	2.9539%	0.8524%	0.8003%	1.9389%	0.8245%	4.7698%	0.2692%	0.9958%
4.6.2010	Crisis in Venezuela	Aggregate demand-side	2.5279%	3.3334%	1.5876%	2.4503%	3.9021%	1.8634%	1.1835%	3.5622%
10.1.2011	Libya's unrest and the subsequent NATO intervention and Saudi Arabia's increase of its oil production	Supply-side	0.2852%	0.9185%	0.4666%	0.5117%	1.4280%	0.4689%	0.0070%	0.1568%
11.3.2011	Japan's Tsunami	Aggregate demand-side	1.2849%	0.2287%	1.4683%	0.5787%	0.3487%	0.6408%	0.9044%	1.3200%
8.8.2011	The downgrade of the U.S. debt status in August 2011	Aggregate demand-side	3.0965%	2.7876%	5.3709%	2.6167%	1.4844%	0.0935%	2.8979%	1.2190%
30.4.2012	The European sovereign debt crisis	Aggregate demand-side	0.0883%	0.4693%	0.2089%	0.0898%	0.3796%	0.1854%	0.1135%	0.1121%
14.7.2014	Robust global production exceeded	Aggregate demand-side	0.0067%	0.0754%	0.2204%	0.2664%	0.0303%	0.0118%	0.1533%	0.0093%
30.6.2015	Greek Debt Crisis Threatens European Union	Aggregate demand-side	3.1997%	6.2871%	2.8642%	3.9143%	1.6524%	0.6793%	0.1391%	0.7126%
23.6.2016	Brexit	Aggregate demand-side	0.4783%	0.5112%	0.3717%	0.4307%	0.1437%	1.2180%	0.7304%	0.4174%
2.12.2016	OPEC and non-OPEC producers reached their first deal since 2001 to curtail oil output jointly (Nearest structural break in crude oil)	Supply-side	0.9427%	1.2592%	0.6505%	8.0926%	1.4373%	1.7057%	3.8208%	0.8143%
7.4.2017	The U.S. missile strike of Syria's Shayrat Airbase	Precautionary demand	0.0647%	0.0508%	0.0552%	0.1958%	0.3943%	0.0052%	0.0277%	0.0063%
22.5.2017	OPEC and non-OPEC members agree to extend production cuts for nine months	Supply-side	0.3785%	0.1080%	0.0242%	0.0222%	0.1740%	0.1739%	0.2168%	0.1112%
6.7.2018	U.S.–China Trade war	Aggregate demand-side	0.7188%	0.0995%	0.2063%	0.0277%	0.0274%	0.5800%	0.0345%	0.0411%
11.3.2020	COVID-19 Pandemic and 2020 Recession	Aggregate demand-side	0.4850%	0.9259%	1.5494%	0.1400%	1.3766%	0.1091%	1.5346%	0.8948%
14	No. of events that have larger impact than trade war		6	11	11	12	13	6	11	11

Notes: This table presents the impact of exogenous shocks on the covariance between crude oil and agricultural commodities. Bold numbers indicate that the event has a higher impact on a particular agricultural product compared to U.S.–China trade war.

4. Discussion of results

We will discuss our results in the context of our three research questions.

RQ1) How does volatility transmit within and between energy (i.e., crude oil) and agricultural (i.e., soybean, corn, wheat, rice, sugar, cotton, coffee, and cocoa) commodity markets according to the VAR-GARCH-BEKK model for the period between July 30, 2007, and April 1, 2021?

Our results reveal that volatilities in crude oil and agricultural commodity markets are affected by market shocks. Specifically, volatilities in the soybean, corn, rice, sugar, cotton, and cocoa markets appear to be affected by shocks from the crude oil market, a finding consistent with Serletis and Xu (2019) who observe unexpected shocks in the impact of oil on the volatility of the sugar market. Lima et al. (2019) similarly correlate changes in the volatility of crude oil with changes in the volatility of sugar, and Umar, Jareno, et al. (2021) show similar results for the impact of shocks from oil prices on volatilities in the cocoa, rice, and sugar markets for the period 2000–2020. Considering a similar period (2002–2020), Umar, Gubareva, et al. (2021) observe the volatilities of cotton, cocoa, soybean, and sugar to be impacted by oil shocks. These findings are partially consistent with Gilbert and Mugera (2014) and Wei and Chen (2016), who identify impacts of crude oil market shocks on corn, soybean, and wheat for the periods 2007–2009 and 2006–2012, respectively. The results from our current study do not observe a significant impact of crude oil volatility on the volatility of wheat.

The volatility of crude oil past returns has been observed to significantly impact the current volatility of corn and cocoa commodities returns, indicating that changes in these agricultural commodity prices are influenced by the volatility of crude oil prices over the sample period (July 30, 2007 to April 1, 2021). Meanwhile, the volatilities of past returns for soybeans and cocoa record significant impacts on the current volatility of crude oil returns. This indicates a bidirectional spillover effect between crude oil and cocoa, a result consistent with the findings of Wu and Li (2013), who report strong evidence of a unidirectional spillover effect from the crude oil market to the corn market for the period between September 2003 and August 2012. The findings are also consistent with those of Dahl et al. (2020), whose results reveal that the static return spillover index for crude oil to corn is higher than that of other agricultural commodities before 2006, and that of crude oil to cocoa is higher than that of other commodities after 2006. Additionally, the static return spillover index for soybeans to crude oil is higher than that of other

agricultural commodities after 2006, and the static return spillover index for cocoa to crude oil is higher than that of the transmission from cocoa to other commodities. This study's findings also support those of Paris (2018), who, while also accounting for biofuel production, observe that oil price does not have a significant long-term impact on wheat prices for the period 2008–2014.

Furthermore, our results are partially consistent with those of Yahya et al. (2019), which indicate that crude oil and soybeans, corn, wheat, sugar, cotton, coffee, and cocoa exhibited significantly higher co-movement during the GFC (2007–2009) and European debt crisis (2010–2012). Notably, Wei and Chen (2016) also conclude that soybeans, corn, and wheat returns are significantly affected by crude oil returns, and Paris (2018) observes a significant oil price effect on the prices of agricultural commodities (corn and soybeans for 2011–2014 and 2008–2014, respectively) due to biofuel development. However, our results conflict with those of Umar, Jareno, et al. (2021), who report that the connection between cocoa and oil prices exhibits a lower transmission level than that of other commodities. This suggests that the results might be affected by the sample size selected for the analysis.

Our results also provide evidence of the asymmetry in the output response to crude oil shocks, suggesting that crude oil and cotton represent variance asymmetry, meaning that a negative price shock will impact price volatility more than a positive price shock (Elder & Serletis, 2009). Similarly, crude oil–soybeans, crude oil–corn, crude oil–coffee, wheat–crude oil and cotton–crude oil exhibit asymmetric cross-market responses, indicating that negative price shock to former commodity increases the volatility of the latter commodity more than positive price shocks of a similar magnitude. Identifying variance asymmetry and volatility spillover could benefit both policymakers and agricultural commodity traders (Shahzad et al., 2018).

RQ2) How does the U.S.–China trade war impact the volatility of crude oil and agricultural commodity markets, and how does it impact covariance between these markets?

Among the selected agricultural commodities, soybeans, coffee, cotton, wheat, and corn demonstrate high levels of responsiveness to the U.S.–China trade war, with the expected volatilities of soybeans (5.32%) most responsive, followed by coffee (4.10%), and cotton (3.55%). Although the U.S.–China trade war also demonstrates a considerable impact on the covariance between markets, the size and dynamic of the shock's impact on the covariance between energy and agricultural commodity markets are commodity-specific. For example, the U.S.–China trade

war represents an enormous impact on the covariances between crude oil and soybeans (0.72%), crude oil and cotton (0.58%), and crude oil and wheat (0.21%), findings that support those of the extant literature and suggest that the tariffs imposed on agricultural commodities affect their price volatility. For instance, soybean prices were likely most affected during this period because soybeans represent a major agricultural commodity in trade between the United States and China. That is, tariffs imposed on soybeans led China to stop purchasing U.S.-produced soybeans in retaliation, likely impacting the commodity's price volatility.

RQ3) How does the impact of the U.S.–China trade war differ from the impact of other exogenous shocks on the variances and covariances of the crude oil and agricultural commodities markets according to the volatility impulse response function approach?

Our results suggest that the price volatility responses of agricultural commodities and the covariance between crude oil and agricultural commodities to exogenous shocks all differ. Notably, the findings reveal that the U.S.–China trade war impacts the variances observed among agricultural commodities more significantly than other exogenous shocks. However, the U.S.–China trade war's impact on the covariances between crude oil and agricultural commodities was smaller than the impact of other exogenous shocks, except in the cases of soybeans and cotton. This suggests that the U.S.–China trade war impacted supply and demand for some agricultural commodities, which supports the statistics indicating that China is the biggest importer of soybeans from the United States (Liu et al., 2018) and the largest importer of cotton⁹, with the total value of cotton exported from the United States to China highest after soybean in 2017 (Liu et al., 2018). Importantly, not only China but also Canada imposed retaliatory tariffs on U.S. coffee (Cadwalader, 2018), further disrupting the supply and demand chains for certain agricultural commodities (i.e., soybeans, cotton, and coffee) as a result of the imposition of tariffs.

Furthermore, the responses of sugar, rice, and cocoa are smaller because the tariffs implemented in the context of the U.S.–China trade war have less impact on the supply and demand situations of these three commodities. The major sugar exporters in the world in 2018–2019 were Brazil and Thailand,¹⁰ with China accounting for 6.4% of the total sugar exports from the United States and

⁹ <https://www.statista.com/statistics/191896/leading-cotton-importing-countries/>

¹⁰ <https://www.statista.com/statistics/273437/exported-amount-of-sugar-in-leading-countries/>

the United States accounting for 10.9% of the total sugar exports from China.¹¹ Meanwhile, the principal rice exporting countries in 2018–2019 were India and Thailand,¹² with the United States only accounting for 2.7% of total rice exports from China during that year.¹³ Finally, the largest importers of cocoa from the United States in 2018 were Canada, Mexico and Korea,¹⁴ with China only accounting for 1.77% of cocoa exports from the United States. Therefore, although China imposed a 25% tariff on U.S. cocoa products,¹⁵ it did not have a substantial overall impact on U.S. cocoa exports. Thus, the results show less pronounced responses of these three agricultural commodities to the U.S.–China trade war compared to their responses to other exogenous shocks.

These findings are consistent with those of Ahmadi et al. (2016), who observe that the response, in terms of volatility, of each commodity to oil price shocks differs significantly depending on the specific underlying cause of the shock. Nazlioglu and Soytas (2012) also confirm that other oil-specific shocks significantly impact agricultural commodity prices through the exchange rate, and Wang et al. (2014) ascertain that both aggregate demand shock (global economic activity) and oil-specific demand shocks (oil market activity) can impact agricultural price volatility. However, Wang et al. (2014) indicate that the explanatory power of oil-specific demand shocks (precautionary demand) on agricultural commodity prices is stronger than that of aggregate demand shocks after 2006. In contrast, our analysis, which includes one oil-specific demand shock, shows that precautionary demand shock has a relatively smaller impact on most agricultural commodities and the covariances between crude oil and agricultural commodities than other types of exogenous shocks.

Furthermore, our results demonstrate differing impacts of shocks on the co-movement from crude oil to agricultural commodities, which are consistent with Su et al. (2019) who claim that the relationship between crude oil and agricultural commodities is unstable over time because of structural breaks, including geopolitical events and crises, and could deliver temporary shocks to the price levels of crude oil and agricultural commodities. Similarly, Al-maadid et al. (2017) observe mixed effects of exogenous shocks, such as food, oil and financial crises, on the dynamic interactions between energy and food prices (namely, oil, ethanol, cocoa, coffee, corn, soybeans,

¹¹ <https://trendeconomy.com/data/h2/UnitedStatesOfAmerica/18>

¹² <https://www.statista.com/statistics/255947/top-rice-exporting-countries-worldwide-2011/>

¹³ <https://trendeconomy.com/data/h2/UnitedStatesOfAmerica/18>

¹⁴ <https://trendeconomy.com/data/h2/UnitedStatesOfAmerica/18>

¹⁵ <https://www.ofimagazine.com/news/cocoa-oil-baked-goods-face-us60bn-tariffs-from-china>

sugar, and wheat), suggesting that policy changes substantially impact the price returns of agricultural commodities and the linkages between price series.

5. Conclusion and policy implications

This study examines the impact of the U.S.–China trade war on energy and agricultural commodity markets using nine-time series data covering Brent crude oil, soybean, corn, wheat, rice, sugar, cotton, coffee, and cocoa over the period July 30, 2007–April 1, 2021. Price volatility behavior has been analyzed using an asymmetric BEKK-GARCH model that contrasts with the impact of the U.S.–China trade war on energy and agricultural commodities, the analysis of which relied on VIRF analysis. This study has also investigated the influences of other exogenous shocks on the volatility of energy and agricultural commodities and the covariance between them, while categorizing the shocks in the spirit of Kilian and Park (2009), and following Eraslan and Ali (2018). The analysis also includes additional exogenous shocks.

The key findings to answer the three research questions are:

- (i) The volatilities of the soybeans, corn, rice, sugar, cotton, and cocoa markets are affected by the shocks from the crude oil market. Significant volatility transmissions are observed for the full sample from crude oil to corn, crude oil to cocoa, soybeans to crude oil, and cocoa to crude oil. There is evidence of asymmetries in terms of volatility patterns, meaning that price declines impact price instability more than price increases. While crude oil–cotton demonstrates variance asymmetry, crude oil–corn, crude oil–sugar, crude oil–coffee, rice–crude oil, cotton–crude oil, and cocoa–crude oil represent cross-price asymmetrical responses.
- (ii) The volatilities of soybeans demonstrate the most responsiveness to the U.S.–China trade war, followed by coffee, cotton, corn, and wheat, with the results suggesting a substantial impact of the U.S.–China trade war on all of the covariances between energy and agricultural commodity markets, and the most significant impacts observed for the covariances between crude oil and soybean, crude oil and cotton, and crude oil and wheat. However, the size and dynamic of the impact of shocks are largely commodity-specific.

- (iii) According to VIRF results, the effect of exogenous shocks on agricultural commodity markets differs depending on the driving forces behind the shocks, with the results indicating that the impact of the U.S.–China trade war on the variances of agricultural commodities, especially soybeans, coffee, cotton, corn and wheat, are more extensive than the impacts of other exogenous shocks. However, the U.S.–China trade war's impact on the covariances between crude oil and agricultural commodities are relatively smaller than the impacts of other exogenous shocks.

Ultimately, these findings provide various insights pertinent to investment decisions, building on confirmation that the structure of volatility patterns might differ according to different exogenous shocks. Otherwise, a lack of knowledge of this structure can make predictions biased and inaccurate. Our results, thus, enable suppliers to adopt more effective strategies, which are likely to enable better inventory planning for agricultural commodities. For example, with the changing environment, suppliers could expand their business by executing their business activities online, such as promoting their products on e-commerce and online platforms, to sustain their business activities. Meanwhile, allowing users to buy agricultural products directly from farmers could reduce overhead costs, and enhanced agricultural R&D could expand the variety and shelf-life of agricultural products.

The spillovers of the U.S.–China trade war have proven critical for commodity-oriented countries, making these results informative for such countries. Therefore, policymakers should consider the relationship between energy and agricultural commodity markets when formulating policies because agricultural commodities are clearly tied to the crude oil market. Furthermore, they should consider the stability of the prices of crude oil and agricultural commodities to minimize the effect of exogenous shocks on prices. Policymakers could also prepare for highly volatile periods by developing appropriate preventive policies. Such policy tools could improve the efficiency of resource allocation by helping producers implement transformational changes in response to exogenous shocks. For example, policymakers could support producers to enhance their manufacturing capabilities by investing in advanced digital production, thereby potentially improving the efficiency of manufacturing production (*Industrial Development Report* 2020).^{16,17}

¹⁶ https://www.unido.org/sites/default/files/files/2020-04/IDR2020_Brief%201_final_0.pdf

¹⁷ https://www.unido.org/sites/default/files/files/2020-07/IDR2020_Brief%204_final.pdf

By building solid production foundations to help producers upgrade their manufacturing capabilities, policymakers could lower the impact of exogenous shocks (*Industrial Development Report* 2020).¹⁸ Furthermore, offering training programs for producers to facilitate a direct experience with, exposure to, and learning from new technologies would encourage and equip workers with advanced digital production skills (*Industrial Development Report* 2020).¹⁹

Our study is not without limitations. It only identifies the responses of commodity volatilities and their volatility linkages to specific shocks at specific times, which might not fully capture and differentiate the impact of different events that occurred during the sample period. This issue can be addressed more explicitly in future studies.

¹⁸ https://www.unido.org/sites/default/files/files/2020-07/IDR2020_Brief%205_final_0.pdf

¹⁹ https://www.unido.org/sites/default/files/files/2020-04/IDR2020_Brief%202_final_0.pdf

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